claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a68c7c8d6b29fb256.jsonl`
- SHA-256: `4f1017544666fae5f54a1c08ff5aa55ceac503c0e198a16e7dd4f739cdd6419f`
- Source modified: `2026-05-31T04:18:11+00:00`
- Imported at: `2026-07-05T16:48:20+00:00`
- project: `wf_076beba8-59c`
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`

Transcript

1. user (2026-05-31T04:17:29.378Z)

Source Extractor

```
Research question: "Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision /
document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud
API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date,
narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST
personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-
column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both
born-digital (text layer present) and the harder cases where pdfplumber's
extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies
between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):
1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default.
Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud
options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT,
BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-
commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This
is critical ? many document-AI models and their training datasets are non-commercial (e.g.
LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several
VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo
license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24
GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license
(state explicitly), latest release / maintenance status as of 2025-2026, VRAM need +
quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL
TABLES / multi-column statements:
- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL
(note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 /
2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community
License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR,
EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND
PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license),
MinerU / PDF-Extract-Kit, RapidOCR.
- Table-structure recognition specifically (cell/grid reconstruction for borderless tables):
TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON
```

transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly."

Fetch and extract key claims from this source:
URL: https://www.bentoml.com/blog/multimodal-ai-a-guide-to-open-source-vision-language-models
Title: Multimodal AI: A Guide to Open-Source Vision Language Models (BentoML)
Found via: Open VLMs for document/table extraction (broad/primary) search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
   - be a concrete, checkable statement (not vague generalities)
   - include a direct quote from the source as support
   - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-05-31T04:17:32.128Z)

I'll start by loading the WebFetch tool schema, then retrieve the page content.

3. assistant (2026-05-31T04:17:35.211Z)

I have the WebFetch tool loaded. Now let me retrieve the BentoML page content.

4. user (2026-05-31T04:17:47.345Z)

Open-Source VLM Claims: Document/OCR/Parsing & Self-Hosted Deployment

VLMs Listed

Models explicitly mentioned:
- GLM-4.6V (106B and 9B Flash variants)
- Gemma 3 (1B, 4B, 12B, 27B sizes)
- Qwen3-VL (235B-A22B and 30B-A3B editions)
- DeepSeek-OCR
- Molmo (1B, 7B, 72B parameters)
- Pixtral (12B parameters)

Related models referenced for comparison:
- Qwen2.5-VL, LLaVa-OneVision, Phi-3.5 Vision, GPT-4V, Claude 3.5 Sonnet

Model Licenses

Pixtral: "released with open weights under the Apache 2.0 license"

No explicit licenses stated for other models, though all are presented as "open-source."

VRAM/Hardware Requirements

Gemma 3: "small sizes make Gemma 3 ideal for deployment in constrained environments such as laptops, desktops, and edge devices"

DeepSeek-OCR speed: "nearly 2,500 tokens per second on a single A100-40G GPU using vLLM"

General note: "Top-performing open-source models...can reach over 70 billion parameters. This means you need high-performance GPUs"

Document/OCR/Table Capabilities

GLM-4.6V: "can convert UI screenshots into clean HTML/CSS/JS with pixel-level accuracy" and handles "multi-document financial reports, research papers, 200-page presentation decks"

Qwen3-VL: "supports OCR in 32 languages" and "accurately parses complex documents, forms, and layouts"

DeepSeek-OCR: "layout analysis, table extraction, chart parsing, chemical formula recognition" and "supports nearly 100 languages"

Molmo: Supports "document reading" and "pointing to objects in images"

Pixtral: Can handle "multi-image conversations" with "128,000 token" context window

Python Integration/Serving Frameworks

vLLM: Mentioned for DeepSeek-OCR deployment

BentoML: "supports a wide range of data types, such as text, images, audios and documents"

OpenLLM: "run `openllm serve pixtral:12b` with OpenLLM"

Publication Date

December 11, 2025

5. assistant (2026-05-31T04:17:54.544Z)

This is a vendor blog (BentoML) from December 11, 2025. It's a secondary source ? a company's marketing/educational blog that surveys open-source VLMs. It provides useful corroborating facts but is not primary research and is somewhat marketing-oriented (BentoML sells serving infrastructure). Let me assess the extracted claims against the research question.

Key relevant claims:
- Pixtral is Apache-2.0 (matches license-safety requirement) ? supporting
- Qwen3-VL OCR/document parsing capability and 32-language OCR ? supporting, but note this is a newer model than the question's named Qwen2/2.5-VL
- DeepSeek-OCR table extraction + throughput on a single A100 with vLLM ? supporting (local single-GPU + Python integration)
- Gemma 3 small sizes for constrained/edge deployment ? tangential
- 70B+ models need high-performance GPUs ? tangential (general VRAM context)

The article does NOT state most licenses (a known gap ? the research question specifically warns weights licenses must be verified). I'll capture that explicitly.

Now I'll submit the structured output.

6. user (2026-05-31T04:18:03.271Z)

Structured output provided successfully

7. assistant (2026-05-31T04:18:11.485Z)

I've completed the analysis of the BentoML article.

Summary: This is a vendor/marketing blog (BentoML, published 2025-12-11) that surveys open-source VLMs and is oriented toward selling serving infrastructure ? so I rated it "blog" quality, useful for corroboration but not primary research.

Key findings relevant to muneem's research question:
- Pixtral 12B is the only model with a confirmed weights license here: Apache-2.0 (commercially safe), servable locally via OpenLLM/vLLM with a 128k context window ? central.
- DeepSeek-OCR does table extraction + layout/chart parsing across ~100 languages and runs on a single A100-40G via vLLM at ~2,500 tok/s ? central, directly matching the single-GPU + Python deployment target.
- Qwen3-VL offers 32-language OCR and document/form/layout parsing (note: this is the newer Qwen3 generation, not the Qwen2/2.5-VL named in the question) ? supporting.
- A general VRAM-tiering note (top models exceed 70B and need high-end GPUs; Gemma 3 1-4B fits edge) ? tangential.
- Important license caveat: the article specifies a weights license ONLY for Pixtral. GLM-4.6V, Qwen3-VL, DeepSeek-OCR, Molmo, and Gemma 3 are merely called "open-source" with no license stated ? directly reinforcing the research question's warning that weights licenses must be independently verified. I flagged this explicitly.

All claims and source-quality rating have been submitted via StructuredOutput.